



Machine Learning-Based Prediction and Classification of Chronic Kidney Disease

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Abstract

Chronic Kidney Disease often progresses silently, making early detection critical to prevent irreversible damage. This study builds and compares machine learning models to classify patients as having chronic kidney disease or not, and identifies the clinical risk factors most associated with the disease. Using a dataset of 400 patients and 25 clinical variables, the data were cleaned, missing values imputed, categorical variables encoded, and split into training and testing sets before balancing the training data using the Synthetic Minority Oversampling Technique. Three algorithms Logistic Regression, Random Forest, and Extreme Gradient Boosting were trained and compared using Accuracy, Precision, Recall, F1 Score, and the Area Under the Receiver Operating Characteristic Curve, while unsupervised K-Means clustering and Principal Component Analysis were also used to reveal latent subgroups beyond the binary diagnosis. All three models achieved comparable performance, each reaching 99.17% accuracy with an Area Under the Receiver Operating Characteristic Curve above 0.99, with hemoglobin, packed cell volume, specific gravity, serum creatinine, and albumin identified as the strongest predictors of the disease, and a three-cluster solution separating patients into CKD (advanced stage), CKD (near-normal stage) and Non-CKD (healthy) groups, supported by the elbow method, silhouette scores, and biomarker patterns, with Principal Component Analysis reinforcing the stability of this structure.

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List of abbreviations

AUC	Area Under the Curve
CKD	Chronic Kidney Disease
LR	Logistic Regression
RF	Random Forest
ROC	Receiver Operating Characteristic
XGBoost	Extreme Gradient Boosting
PCA	Principal Component Analysis
WCSS	Within-Cluster Sum of Squares

Chapter 1

Introduction

Chronic Kidney Disease (CKD) is one of the most common long-term health conditions worldwide, and it often develops without early warning signs. This chapter introduces the problem addressed in this study, explains its importance, and outlines the objectives and structure of the research.

1.1 Research problem

CKD often shows no clear symptoms in its early stages, so many patients are diagnosed only after significant kidney damage has occurred. Diagnosis relies on interpreting multiple clinical measurements together, which can be difficult and time-consuming. In addition, it remains uncertain which machine learning algorithm provides the most reliable predictive performance for this type of medical data and which clinical variables contribute most to accurate disease prediction. This study addresses these challenges by building and comparing three classification algorithms—Logistic Regression, Random Forest, and Extreme Gradient Boosting (XGBoost)—to classify patients as CKD or non-CKD and identify the variables most strongly associated with the disease.

1.2 Research Importance

Early detection of CKD is important because it allows treatment to begin before the disease reaches a more advanced and harder-to-treat stage. A reliable classification model can help physicians reach quicker and more consistent decisions, especially in settings where specialist resources are limited. Identifying the most important risk factors is equally valuable, since it helps prioritize which clinical tests and measurements are most useful for detecting CKD early, rather than relying on the full and sometimes costly set of tests. Furthermore, comparing multiple machine learning algorithms on the same dataset provides useful guidance on which type of model is best suited for this kind of medical, tabular data — knowledge that can be applied to similar disease-prediction problems in the future.

1.3 Research objectives

The objectives of this study are:

1. To classify patients as CKD or not-CKD using three supervised classification algorithms: Logistic Regression, Random Forest, and XGBoost.
2. To compare the performance of the three classification algorithms on an identical train-test split, using a unified set of evaluation metrics: Accuracy, and ROC-AUC.
3. To identify the clinical variables most strongly associated with CKD, using Random Forest variable importance (Mean Decrease Gini and Mean Decrease Accuracy).
4. To explore whether latent subgroups exist among patients, beyond the simple CKD / not-CKD classification, using K-Means clustering and applying Principal Component Analysis to uncover underlying structure in the data.

1.4 Definitions and Concepts

- **Chronic Kidney Disease (CKD):** A long-term condition in which the kidneys are damaged and gradually lose their ability to filter waste from the blood.
- **Machine Learning:** A branch of artificial intelligence in which a computer learns patterns from data, rather than following manually written rules.
- **Supervised Classification:** A type of machine learning task in which a model learns from data where the correct answer (label) is already known, in order to predict the label of new, unseen data.
- **Risk Factor:** A clinical variable that is statistically associated with an increased likelihood of a disease being present.
- **Clustering:** Unsupervised learning refers to methods that analyze data without predefined labels to uncover hidden patterns. Clustering is an unsupervised technique that groups similar observations based on shared characteristics, allowing the discovery of natural subgroups within the dataset.
- **Principal Component Analysis:** A dimensionality-reduction technique that transforms a large set of correlated variables into a smaller set of uncorrelated components (principal components), while preserving as much of the original variance as possible.

1.5 Outline of the Research

The remainder of this report is organized as follows. Chapter 2 reviews previous studies related to machine learning approaches for CKD prediction. Chapter 3 describes the dataset and the methodology used in this study, including data preprocessing, the classification and clustering algorithms, and the evaluation techniques. Chapter 4 presents the results of the classification models and the

clustering analysis. Chapter 5 summarizes the main findings, discusses their implications, and suggests directions for future research.

Chapter 2

Literature Review

2.1 Introduction

This chapter reviews some empirical studies relevant to the prediction and classification of chronic kidney disease. Since the present research aims to classify patients as CKD / not-CKD using Logistic Regression, Random Forest, and XGBoost, the studies are organized around three complementary strands of the literature that map directly onto the objectives of this study: (a) studies that directly compare multiple classification algorithms on CKD datasets, supporting the first objective on classifier performance; (b) studies that integrate feature selection, clustering, or class-imbalance handling into the modelling pipeline, supporting the second objective on preprocessing methodology; and (c) broader validation and alternative-data studies that contextualize the reliability and generalizability of CKD prediction models, supporting the third objective.

2.2 Comparative Studies of Classification Algorithms for CKD Prediction

(Islam et al., 2023) Investigated CKD prediction using a 25-variable clinical dataset, evaluating twelve supervised classifiers including Logistic Regression, Random Forest, Decision Tree, SVM, KNN, Naïve Bayes, ANN, AdaBoost, Gradient Boosting, Extra Trees, SGD, and XGBoost following preprocessing and feature selection. The findings showed that XGBoost achieved the highest performance (98.3% accuracy, 98% precision, recall, and F1-score), and the study concluded that XGBoost can effectively support early CKD diagnosis and clinical decision-making.

(Chittora et al., 2021) Examined CKD prediction on the UCI dataset using seven classifiers (ANN, C5.0, CHAID, Logistic Regression, Linear SVM with L1 and L2 penalties, and Random Tree), each evaluated under four feature selection strategies: full feature set, correlation-based selection, wrapper method, and LASSO regression. The findings provided a comprehensive comparison of classifier performance across these feature selection strategies, and the study concluded that combining suitable feature selection with machine learning classifiers enhances the diagnostic accuracy of CKD prediction models.

(Debal and Sitote, 2022) investigated CKD prediction and staging using a clinical dataset of 1,718 patient records from St. Paul’s Hospital, Ethiopia, covering both binary (CKD/non-CKD) and multiclass (disease stage) classification. Feature selection was performed using ANOVA and Recursive Feature Elimination with Cross-Validation before training Random Forest, SVM, and Decision Tree classifiers under 10-fold cross-validation. The findings showed that Random Forest achieved the highest performance in both binary and multiclass classification, and the study concluded that Random Forest is a reliable decision-support tool for CKD prediction and staging in clinical practice.

(Khan et al., 2024) evaluated Logistic Regression, Random Forest, Decision Tree,

KNN, and SVM (with linear, Laplacian, Bessel, and radial basis kernels) using case-control data from CKD patients in Buner district, Khyber Pakhtunkhwa, Pakistan, comparing models via accuracy, Brier score, sensitivity, Youden’s index, and F1-score. The findings showed that ensemble and kernel-based models consistently outperformed simpler statistical methods, and the study concluded that machine learning can meaningfully support early CKD diagnosis, suggesting the approach could extend to other diseases such as heart disease and cancer.

(Sharma et al., 2025) developed a CKD detection model, comparing nine classifiers (Extra Trees, XGBoost, Decision Tree, Gradient Boosting, Stochastic Gradient Boosting, CatBoost, AdaBoost, Random Forest, and KNN) after preprocessing and correlation analysis. The findings showed that Extra Trees and XGBoost achieved the highest accuracy (99.16% each), with most tree-based ensembles exceeding 98% while KNN performed weakest (63.33%), and hemoglobin and packed cell volume showed the strongest correlations with CKD status. The study concluded that ensemble-based classifiers substantially outperform simpler models such as KNN, and recommended integrating such models into web-based screening tools.

2.3 Studies Integrating Feature Selection, Clustering, and Imbalance Handling

(Hassan et al., 2023) conducted a study on the UCI CKD dataset , after which K-Means clustering was applied to identify latent patient groups (three clusters, via the elbow method), and XGBoost with SHAP was used to reduce the 25 features to the 7 most important, led by hemoglobin and specific gravity. Five classifiers were trained on both feature sets; the Neural Network achieved the highest accuracy (100%) on the full dataset, while SVM reached 100% on the reduced set, and Random Forest remained stable at 98.75% on both. The study concluded that combining clustering with XGBoost-based feature selection im-

proves or maintains performance while reducing dimensionality.

(Imaduddin et al., 2026) compared SMOTE and cost-sensitive learning across six classifiers (Decision Tree, KNN, Logistic Regression, Random Forest, SVM, and XGBoost) on a CKD dataset of 1,659 records with severe class imbalance (1,524 versus 135), using Random Forest feature importance to select the top 20 of 54 variables, led by serum creatinine and GFR. The findings showed that cost-sensitive learning generally improved precision while SMOTE improved recall and F1-score, with the best result achieved by XGBoost with cost-sensitive learning (93% accuracy, 92% precision) and Random Forest the most stable classifier overall. The study concluded that the choice of imbalance-handling technique should depend on the specific classifier rather than being applied uniformly.

(Yang et al., 2024) applied eleven algorithms — eight traditional classifiers and three neural models (ANN, RNN, LSTM) — to , testing unbalanced, SMOTENC-balanced, and feature-selected (14 features) versions under 10-fold cross-validation. Accuracies ranged from 94% to 99.4%, with Random Forest, SVM, AdaBoost, and XGBoost combining strong accuracy with low computation time, while statistical testing showed no significant difference between the balanced, unbalanced, and feature-selected datasets, and deep learning models showed no accuracy advantage despite longer training times. The study concluded that SMOTE offers no statistically significant benefit for a dataset this size, favoring Random Forest, SVM, AdaBoost, and XGBoost as the most practical classifiers.

(Chhabra et al., 2024) compared classification algorithms on the full- feature UCI CKD dataset, applying SMOTE to address class imbalance and k-fold cross-validation for evaluation, then combined the top three performers — SVM, Random Forest, and AdaBoost — into a stacking ensemble. The findings showed that the stacking ensemble achieved the highest accuracy (99.5%), outperforming all individual classifiers, and the study concluded that combining SMOTE-based balancing with ensemble stacking significantly improves the accuracy and robustness of CKD prediction, suggesting the approach could extend to other disease-prediction

tasks.

2.4 Broader Validation, Statistical, and Alternative-Data Approaches

(Lim et al., 2022)conducted a scoping review screening 516 studies and including 33 for full-text analysis of predictive models for CKD progression, examining the statistical methods, clinical variables, and performance typically reported. The findings showed that logistic regression was the most frequently applied technique, with Cox proportional hazards regression common for time-to-progression studies, and that serum creatinine, blood urea, albuminuria, specific gravity, hemoglobin, hypertension, and diabetes were the strongest recurring predictors. The study concluded that traditional statistical models remain valuable and interpretable in clinical settings, though stronger external validation and comparison with machine learning approaches are still needed.

(Zhang et al., 2026), externally validated 16 previously published CKD and kidney-failure risk prediction models using data from 497,896 UK Biobank participants, assessing discrimination (c-index) and calibration across demographic and clinical subgroups. The findings showed fair-to-excellent discrimination for CKD (c-indices 0.60–0.81), but models frequently overpredicted CKD risk while underpredicting kidney-failure risk, with the O’Searghdha model showing the best overall performance. The study concluded that most existing CKD risk models require recalibration before clinical use, and that discrimination was generally poorer among people with diabetes or hypertension — a caution relevant to generalizing any single-dataset model

(Mendapara, 2024)developed a CKD prediction model using gene expression data from the Gene Expression Omnibus database, applying Lasso combined with Random Forest to isolate six genetic biomarkers before training a final Random Forest classifier, validated with 10-fold cross-validation and an independent external dataset. The findings showed strong predictive performance (91.18% accuracy, AUC 0.979 internally; 89.83% accuracy, AUC 0.913 externally). The study concluded that the identified biomarkers and Random Forest model show

strong potential for early CKD detection, though reliance on gene expression data alone may limit applicability to routine clinical settings.

Chapter 3

Methodology

3.1 Introduction

This chapter explains how the data was collected, cleaned, and prepared for analysis. It also describes the machine learning methods used to classify patients as CKD or not-CKD, and the measures used to evaluate and compare the models.

3.2 Data

The dataset was obtained from Kaggle (2017), and was originally collected from a hospital in Tamil Nadu, India by Rubini et al., 2015 . It contains 400 patients described by 25 variables: 24 predictors and 1 target variable.

3.3 Data Description

The dataset has 25 variables in total: 24 predictor variables and 1 target variable. Table 3.1 lists each variable, a brief description, type, and possible values or unit of measurement.

Table 3.1: Description of the dataset variables

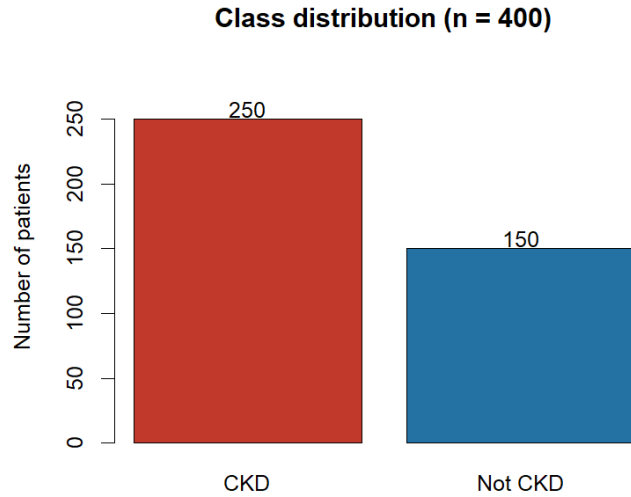
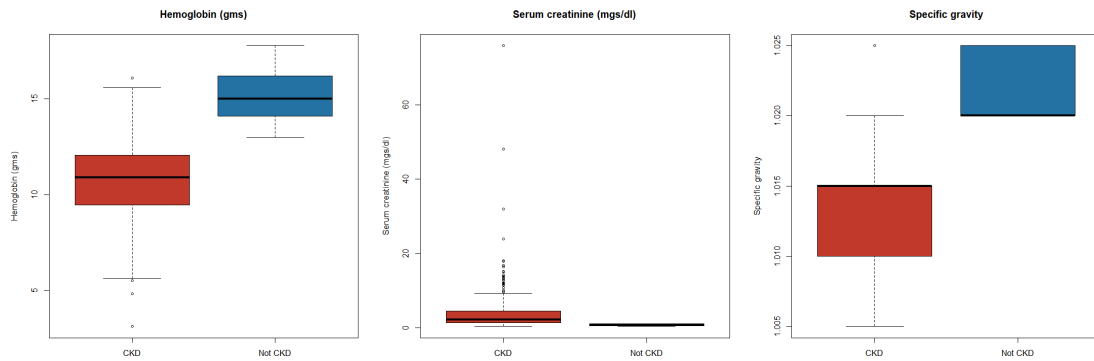
Variable	Description	Type	Values / Unit
age	Age	Numeric	years
bp	Blood pressure	Numeric	mm/Hg
sg	Specific gravity	Categorical	1.005, 1.010, 1.015, 1.020, 1.025
al	Albumin	Categorical	0–5
su	Sugar	Categorical	0–5
rbc	Red blood cells	Categorical	normal, abnormal
pc	Pus cell	Categorical	normal, abnormal
pcc	Pus cell clumps	Categorical	present, notpresent
ba	Bacteria	Categorical	present, notpresent
bgr	Blood glucose random	Numeric	mgs/dl
bu	Blood urea	Numeric	mgs/dl
sc	Serum creatinine	Numeric	mgs/dl
sod	Sodium	Numeric	mEq/L
pot	Potassium	Numeric	mEq/L
hemo	Hemoglobin	Numeric	gms
pcv	Packed cell volume	Numeric	%
wc	White blood cell count	Numeric	cells/cumm
rc	Red blood cell count	Numeric	millions/cmm
htn	Hypertension	Categorical	yes, no
dm	Diabetes mellitus	Categorical	yes, no
cad	Coronary artery disease	Categorical	yes, no
appet	Appetite	Categorical	good, poor
pe	Pedal edema	Categorical	yes, no
ane	Anemia	Categorical	yes, no
classification	Target: CKD status	Categorical (target)	ckd, notckd

Out of the 24 predictor variables, 11 are numeric, and 13 are categorical, in addition to the categorical target variable. Several of these variables — such as specific gravity (**sg**), hemoglobin (**hemo**), serum creatinine (**sc**), packed cell volume (**pcv**), and albumin (**al**) — are commonly reported in the CKD literature as strong predictors of disease status, which is consistent with the risk-factor findings discussed later in this study.

To further characterize the dataset, descriptive statistics were computed for each numeric variable on the raw data prior to cleaning, as summarized in Table 3.2.

Table 3.2: Descriptive statistics of the numeric variables (raw data)

Variable	Mean	Variance	Min	Max	Missing
Age (years)	51.48	294.80	2.00	90.00	9
Blood pressure (mm/Hg)	76.47	187.24	50.00	180.00	12
Specific gravity	1.02	0.00	1.00	1.02	47
Albumin	1.02	1.83	0.00	5.00	46
Sugar	0.45	1.21	0.00	5.00	49
Blood glucose random (mgs/dl)	148.04	6285.59	22.00	490.00	44
Blood urea (mgs/dl)	57.43	2550.55	1.50	391.00	19
Serum creatinine (mgs/dl)	3.07	32.96	0.40	76.00	17
Sodium (mEq/L)	137.53	108.34	4.50	163.00	87
Potassium (mEq/L)	4.63	10.20	2.50	47.00	88
Hemoglobin (gms)	12.53	8.48	3.10	17.80	52
Packed cell volume (%)	38.88	80.82	9.00	54.00	71
White blood cell count (cells/cumm)	8,406.1	8,669,928.3	2,200	26,400	106
Red blood cell count (millions/cmm)	4.71	1.05	2.10	8.00	131

**Figure 3.1:** Distribution of CKD and not-CKD patients in the dataset**Figure 3.2:** Distribution of hemoglobin, serum creatinine, and specific gravity by CKD status

3.4 Data Preprocessing

Before the three models were built, the data set was prepared using five steps. These steps were necessary because the raw data contained errors, missing values, and some variables stored in an incorrect data type.

3.4.1 Data Cleaning

The ID column was removed first because it is just a row number and does not help in prediction. Some categorical values had extra spaces or tab characters in them, so these were removed. One value in the target column was written as "ckd" instead of "ckd", so this was fixed to match the correct label. Also, three variables — pcv, wc, and rc — were stored as text instead of numbers because of some invalid characters in the data. These three variables were changed back to numeric type so they could be used in the models.

3.4.2 Missing Values

Several variables in the dataset contained missing values. For numeric variables like age, hemoglobin, and serum creatinine, the missing values were replaced with the median. The median was used instead of the mean because it is less affected by outliers. For categorical variables like hypertension, diabetes mellitus, and anemia, the missing values were replaced with the mode, which is the most common category for that variable.

3.4.3 Encoding

All categorical variables were converted into numeric or binary values so the models could use them. The target variable, classification, was also converted into a binary variable with two levels: ckd and notckd.

3.4.4 Splitting the Data

The dataset was split into training and testing sets first, before any balancing was done. This used a stratified 70:30 split on the original, imbalanced data. The split kept the same class proportions as the original dataset (62.5% `ckd`, 37.5% `notckd`) in both sets. This gave 280 training observations (175 `ckd`, 105 `notckd`) and 120 test observations (75 `ckd`, 45 `notckd`). Splitting the data before balancing means the test set contains only real, unmodified patient records. This gives a more realistic evaluation of how the model performs on new data.

3.4.5 Class Imbalance

The training set had a class imbalance: 175 `ckd` cases and 105 `notckd` cases. To fix this, the Synthetic Minority Oversampling Technique (SMOTE)

(Chawla et al., 2002) was applied to the training set only. It created synthetic samples for the minority class (`notckd`), increasing it from 105 to 175 cases. This made the training set balanced, with 350 total observations: 175 in each class. The test set was not touched by this step. It kept its original, imbalanced distribution, so it still reflects the real-world proportions of CKD patients.

SMOTE creates synthetic samples based on the nearest neighbors of real data points. If SMOTE is applied before splitting the data, some synthetic samples in the test set could be based on real samples from the training set, or the other way around. This is a known problem in machine learning called data leakage

(Lemaître et al., 2017). It makes the model’s performance look better than it really is. To avoid this, the data was split first, and SMOTE was applied only to the training set. This way, the test set contains only real, unseen patient records, which gives a more accurate picture of how the model would perform in practice.

3.4.6 Standardization

In this study, standardization was applied exclusively to the variables used in the K-Means clustering and PCA analyses, as these methods rely on distance-based and variance-based computations that require all features to operate on a comparable numerical scale.

(Jolliffe and Cadima, 2016) The classification models used earlier did not require this step, but clustering and PCA depend critically on feature scaling to ensure valid and interpretable results. Standardization was applied as a core preprocessing step to ensure that all continuous variables contributed equally to the clustering and dimensionality reduction procedures. The CKD dataset contains biochemical measurements that vary substantially in scale and units—for example, serum creatinine (sc) may reach values above 10 mg/dL in severe cases, whereas specific gravity (sg) typically ranges between 1.01 and 1.02. Without standardization, variables with larger numeric ranges would dominate Euclidean distance calculations, biasing the clustering results and masking clinically meaningful patterns.

3.5 Classification Methods

Machine learning is a branch of artificial intelligence that allows a computer to learn patterns from data, instead of following rules written manually by a person. In this study, machine learning is used to build a model that can predict whether a patient has chronic kidney disease (CKD) or not, based on the 25 clinical variables described in Section 3.3.

Classification is a supervised learning task in which the goal is to predict a qualitative, categorical response variable, in contrast to regression, which predicts a quantitative response (Hastie et al., 2009). In this study, each patient is classified into one of two classes: `ckd` or `notckd`. When the model is trained using data

where the correct class of each observation is already known, this is called supervised classification, since the model “learns” from labeled examples before being used to predict the class of new, unseen patients. Three classification algorithms were used and compared in this study: Logistic Regression, Random Forest, and XGBoost. Each one is explained in the following subsections.

3.5.1 Random Forest

Random Forest (RF) (Breiman, 2001) is an ensemble learning method that builds many decision trees and combines their predictions into a single, more stable classifier. Instead of training one tree on the full dataset, each tree is trained on a different bootstrap sample — a random sample of the training data drawn with replacement, of the same size as the original training set. Figure 3.3 illustrates this process using a simplified example with three trees: three bootstrap samples are drawn from the training data, each tree is trained on its own sub-dataset, and each tree produces its own prediction(Breiman, 2001)



Figure 3.3: RF classification process for a simplified example with three trees

On average, each bootstrap sample leaves out about 37% of the original observations. These left-out cases, known as out-of-bag (OOB) observations, can later be used to estimate the model’s error without needing a separate validation set. A second source of randomness is what sets RF apart from simple bagging. At every split within every tree, the algorithm considers only a random subset of m predictors, rather than all of the available predictors. For classification problems, m is typically set to the square root of the total number of predictors. Restricting the candidate predictors at each split prevents a few dominant variables from controlling every tree, which makes the trees less similar to one another and further reduces the variance of the final ensemble compared to bagging alone.

Among the candidate predictors considered at a split, the one chosen is whichever produces the greatest improvement in node purity, measured using the Gini index:

$$Gini = 1 - \sum_{k=1}^K p_k^2$$

where p_k is the proportion of observations belonging to class k within that node, and K is the number of classes (here, $K = 2$: `ckd` and `notckd`). Each tree is grown until a stopping criterion is reached, without pruning. Because Random Forest does not become more prone to overfitting as more trees are added, increasing the number of trees B mainly affects computation time rather than predictive performance.

Once all trees are built, a new patient is classified by majority vote, as shown in Figure 3.1: each tree casts a vote for a class (`ckd` or `notckd`), and the class receiving the most votes becomes the forest’s final prediction.

Beyond classification, Random Forest also provides a way to rank the importance of each predictor, which supports the second objective of this study — identifying the strongest clinical risk factors for CKD. Two importance measures were used: Mean Decrease Gini, and Mean Decrease Accuracy (permutation importance), which measures the drop in out-of-bag prediction accuracy when a variable’s values are randomly shuffled, breaking its relationship with the outcome (Breiman,

2001). Formally, the Mean Decrease Gini importance of a predictor $X^{(j)}$ is defined as

$$\widehat{\text{MDI}}(X^{(j)}) = \frac{1}{B} \sum_{b=1}^B \sum_{t: j_{n,t}^* = j} p_{n,t} L(j_{n,t}^*, z_{n,t}^*)$$

where B is the number of trees in the forest, the inner sum runs over all nodes t in tree b where predictor j was selected for splitting, $p_{n,t}$ is the proportion of observations reaching node t , $j_{n,t}^*$ denotes the variable chosen for splitting at node t , and $z_{n,t}^*$ denotes the corresponding split threshold at that node. $L(\cdot)$ is the resulting decrease in the Gini index from that split. Variables associated with a larger decrease in either measure are considered more influential in distinguishing CKD from non-CKD patients.

3.5.2 Logistic Regression

Logistic Regression(LR) is a statistical technique widely used for classification problems in which the dependent variable is categorical rather than continuous. Instead of modeling the response directly, the method estimates the probability that an observation belongs to a particular class based on a set of independent variables, which makes it especially suited to binary outcomes such as Yes/No, Positive/Negative, or Disease/Non-disease(Hastie et al., 2009).

The model relies on the logistic (sigmoid) function, which compresses a linear combination of the predictor variables into a probability bounded between 0 and 1:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p)}}$$

where:

- $P(Y = 1 \mid X)$: Probability that the observation belongs to the positive

class.

- Y : Binary response variable ($1 = \text{CKD}$, $0 = \text{Non-CKD}$).
- $X_1, X_2, \dots, X_p$: Predictor (independent) variables.
- β_0 : Intercept of the model.
- $\beta_1, \beta_2, \dots, \beta_p$: Regression coefficients corresponding to each predictor.
- e : Base of the natural logarithm.

An equivalent way to express the model is through the logit function, which is linear in the predictors:

$$\ln \left(\frac{P(Y = 1)}{1 - P(Y = 1)} \right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$$

where:

- $\ln \left(\frac{P(Y=1)}{1-P(Y=1)} \right)$: Log-odds of the positive class.
- $P(Y = 1)$: Probability that the patient has CKD.
- β_0 : Intercept.
- $\beta_1, \beta_2, \dots, \beta_p$: Regression coefficients.
- $X_1, X_2, \dots, X_p$: Predictor variables.

This form explains why LR can be fitted using linear-model machinery even though the predicted probability itself is nonlinear and bounded.

To convert an estimated probability into a class label, a decision threshold is applied, conventionally 0.5:

$$\hat{Y} = \begin{cases} 1, & P(Y = 1) \geq 0.5 \\ 0, & P(Y = 1) < 0.5 \end{cases}$$

where:

- $\hat{Y}$: Predicted class.
- $P(Y = 1)$: Predicted probability of CKD.
- 1: Classified as CKD.
- 0: Classified as Non-CKD.

Model coefficients are estimated using Maximum Likelihood Estimation (MLE) rather than Ordinary Least Squares (OLS). The log-likelihood function is given by:

$$\ell(\beta) = \sum_{i=1}^n [y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i)]$$

where:

- $\ell(\beta)$: Log-likelihood function.
- n : Number of observations.
- y_i : Observed class label of observation i .
- p_i : Predicted probability of observation i .
- $\beta = (\beta_0, \beta_1, \dots, \beta_p)$: Vector of regression coefficients.

Since this function has no closed-form solution, the coefficients are estimated iteratively using optimization algorithms such as Newton–Raphson.

A major advantage of LR is that its coefficients remain interpretable. Each coefficient represents the change in the log-odds of the outcome associated with a one-unit increase in the corresponding predictor while holding all other variables constant. Exponentiating the coefficient, e^{β_i} , gives the odds ratio, which provides an intuitive measure of the association between each predictor and the outcome. This interpretability makes LR particularly valuable in medical research, where identifying influential risk factors is often as important as achieving accurate predictions (Dreiseitl and Ohno-Machado, 2002; Sperandei, 2014).

3.5.3 XGBoost

XGBoost is a supervised learning algorithm that builds a classification model using many decision trees, similar to Random Forest. However, unlike Random Forest, where trees are built independently of each other, XGBoost builds trees one at a time, and each new tree is trained to correct the mistakes made by the trees built before it (Chen & Guestrin, 2016). This step-by-step correction process is called boosting.

XGBoost improves its trees using an optimization method called gradient descent, minimizing an objective function made up of two parts: a term that measures how well the model fits the training data, and a regularization term that discourages the model from becoming too complex (Chen & Guestrin, 2016). Formally, the objective function at each boosting round is defined as:

$$Obj = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)$$

where n is the number of patients in the training set and K is the total number of trees in the model. $l(y_i, \hat{y}_i)$ measures the difference between the true label y_i and the predicted value $\hat{y}_i$ for each patient, and $\Omega(f_k)$ penalizes the complexity of each tree f_k , defined as:

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2$$

where T is the number of leaves in the tree and w represents the leaf weights. This regularization term is what helps prevent overfitting.

For classification problems like this one, the combined output of all the trees is passed through a sigmoid function, converting it into a probability between 0 and 1. A threshold of 0.5 is then applied: patients with a predicted probability of 0.5 or higher are classified as `ckd`, and the rest as `notckd` (Chen & Guestrin, 2016)

XGBoost was trained using the following configuration:

- Objective: `binary:logistic`
- Evaluation metric: log-loss
- Maximum tree depth: 4
- Learning rate (η): 0.1
- Regularization parameters: $\gamma = 1$, $\lambda = 1$
- Number of boosting rounds: 100, with early stopping applied if no improvement was observed after 10 consecutive rounds

3.6 Performance Evaluation Metrics

After training each model, its performance was evaluated on the test set using the following metrics, allowing a fair and unified comparison between Logistic Regression, Random Forest, and XGBoost.

3.6.1 Confusion Matrix

A confusion matrix summarizes the model’s predictions by comparing them to the true class labels. It has four possible outcomes: True Positives (TP), patients correctly predicted as `ckd`; True Negatives (TN), patients correctly predicted as `notckd`; False Positives (FP), patients incorrectly predicted as `ckd`; and False Negatives (FN), patients incorrectly predicted as `notckd`. All other metrics in this section are calculated from these four values.

3.6.2 Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

3.6.3 Precision

$$Precision = \frac{TP}{TP + FP}$$

3.6.4 Recall (Sensitivity)

$$Recall = \frac{TP}{TP + FN}$$

3.6.5 F1-score

The F1-score combines Precision and Recall into a single value, using their harmonic mean (Powers, 2011):

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

3.6.6 ROC Curve and AUC

The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate (Recall) against the False Positive Rate at different classification thresholds, showing how the model's performance changes as the threshold for predicting `ckd` is varied. The Area Under the Curve (AUC) summarizes the ROC curve into a single number between 0 and 1, representing the model's overall ability to distinguish between `ckd` and `notckd` patients. An AUC of 1 indicates perfect separation between the two classes, while an AUC of 0.5 indicates performance no better than random guessing (Fawcett, 2006).

3.7 Clustering

Clustering is an unsupervised learning method that groups observations based on similarity without using predefined labels. Its purpose is to uncover natural structure within the dataset and identify meaningful subgroups that emerge from the data itself.

3.7.1 Standardization

Standardization converts a variable X with mean μ and standard deviation σ into a dimensionless standard score Z using the transformation, (Montgomery and Runger, 2014):

$$Z = \frac{X - \mu}{\sigma}$$

In this process, mean centring shifts the distribution's mean to zero by subtracting μ from each observation, while scaling divides by σ to rescale the distribution's variance and standard deviation to 1.

Consequently, every standardized value Z represents the precise distance between an observation X and the mean μ in units of standard deviation, allowing direct mathematical comparison across variables measured on completely different scales.

3.7.2 K-Means Clustering

K-Means is a distance-based clustering algorithm used to partition the dataset into k groups based on similarity in feature space. Its role in this study is to uncover the underlying CKD severity structure without relying on predefined labels. After representing each observation as a numerical vector, the algorithm iteratively assigns points to the nearest centroid and updates cluster centers until convergence. This makes K-Means suitable for identifying natural patient subgroups that emerge directly from the biochemical and clinical measurement

The process begins by representing the dataset as numerical vectors. $x_i \in \mathbb{R}^d$...such that each sample represents a set of features. The process begins by representing the dataset as numerical vectors $x_i \in \mathbb{R}^d$ such that each sample represents a set of features:

$$X = \{x_1, x_2, \dots, x_n\}, \quad x_i \in \mathbb{R}^d$$

The clustering technique aims to divide data into internally homogeneous but externally distinct groups without pre-defined labels:

$$C_1, C_2, \dots, C_k$$

It consists of n samples with d features and is divided into K groups such that the points within each group are close to one another according to a standard distance function, such as Euclidean distance, (Bishop, 2006):

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^d (x_{ik} - x_{jk})^2}$$

It is assumed that the data can be partitioned into K groups such that the points within each group are as close together as possible, while the groups themselves are far apart from one another. To achieve this objective, the K-means algorithm was employed; this algorithm relies on the pre-selection of the number of clusters, k . The algorithm begins by selecting initial cluster centers, then assigns each point to the nearest center based on a distance function, thereby forming initial clusters. Subsequently, the cluster centres are updated by calculating the arithmetic mean of all points belonging to each cluster, according to the relationship:

$$\mu_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$$

The assignment and update process repeats until a stable state is reached where the cluster centers do not change significantly, indicating that the algorithm has arrived at a suitable solution.

3.7.3 Within-Cluster Sum of Squares

To measure the quality of this clustering, the WCSS was calculated; this metric measures the degree of dispersion within each cluster and is defined as the sum

of squared distances between each point and its cluster's centroid:

$$\text{WCSS}_K = \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

A lower WCSS value indicates greater separation between the clusters. However, selecting the number of clusters k poses a methodological challenge; the Elbow Method is employed to determine this number. This method involves running the K-means algorithm for various values of k , calculating the WCSS for each, and plotting the relationship between k and WCSS. The "elbow point" is identified as the point where the rate of WCSS decrease begins to level off (Ketchen and Shook, 1996), indicating that increasing the number of clusters beyond this point does not significantly improve the quality of the fit.

3.7.4 Silhouette Score

After determining the optimal number of clusters, the quality of the clustering was evaluated using the Silhouette Score, which measures how close each point is to its own cluster compared to other clusters. This score is calculated by determining the average distance between a point and the other points within its cluster:

$$a(i) = \frac{1}{|C_i| - 1} \sum_{x_j \in C_i, j \neq i} d(x_i, x_j)$$

Then, the average distance between the point and the nearest other group is calculated:

$$b(i) = \min_{k \neq C_i} \frac{1}{|C_k|} \sum_{x_j \in C_k} d(x_i, x_j)$$

The silhouette index for a point is defined by the relationship, (Rousseeuw, 1987):

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

This formulation ensures that:

- $s(i) \approx 1$: The observation is well matched to its own cluster and poorly matched to neighbouring clusters.
- $s(i) \approx 0$: The observation lies near the boundary between two clusters.
- $s(i) < 0$: The observation may be incorrectly assigned, being closer to another cluster than its own.

The overall average of the silhouette values for all points was calculated to assess the quality of the clustering as a whole, and this metric was used to verify that the chosen number of clusters achieves an appropriate balance between cluster cohesion and separation.

3.7.5 Principal Component Analysis

PCA is a mathematical methodology aimed at dimensionality reduction while preserving as much data variance as possible. The process begins by normalizing numerical features to ensure all variables operate within a unified numerical range—a necessary step because PCA relies on variance analysis, and significant disparities in value ranges could bias the components toward variables with higher variance. Following normalization, the covariance matrix is calculated (Jolliffe, 2002):

$$\Sigma = \frac{1}{n-1} X^T X$$

Next, the eigenvalues and eigenvectors of this matrix are extracted; the eigenvalues represent the amount of variance explained by each principal component, while the eigenvectors represent the new directions onto which the data is projected. An appropriate number of principal components is selected based on the amount of explained variance, retaining those components that account for the majority of the variance; the data is then re-represented in this new space according to the relationship:

$$Z = XW_K$$

where W_K is the matrix of selected eigenvectors. At this stage, the data is represented as principal components of lower dimensionality that are better able to capture the data's internal structure compared to the original variables, which may contain noise or uninformative linear correlations. After obtaining the data in the PCA space, these components were used as inputs for the K-means algorithm instead of the original variables. This combination of PCA and K-means is a common statistical approach because it mitigates the impact of insignificant features, ensuring the clustering process relies on the data's true structure rather than noise. Furthermore, dimensionality reduction enhances K-means performance; since the algorithm relies on distance calculations, fewer dimensions result in more stable distances that are less susceptible to distortions.

Subsequently, the clustering was evaluated using the WCSS criterion to assess cluster homogeneity; the Elbow Method was employed to determine the optimal number of clusters; and finally, the Silhouette Score was used to verify clustering quality within the new space resulting from PCA. The results demonstrated that using principal components instead of the original variables led to a marked improvement in cluster separation and homogeneity, reflecting the effectiveness of combining PCA and K-means in revealing the data's internal structure (Ding and He, 2004)

Chapter 4

Results and Discussion

4.1 Introduction

This chapter presents the results obtained from applying the methodology described in Chapter 3. It first reports the performance of the three classification algorithms LR, RF, and XGBoost on the test set. The most influential clinical variables identified through RF are then discussed. The chapter also presents the results of the K-Means clustering analysis, which explored whether patient subgroups exist beyond the binary CKD / not-CKD classification. Finally, the performance of the three classification models is compared directly to determine which algorithm is best suited to this type of medical data .

4.2 Data Preprocessing Results

summarizes the missing values across the numeric variables prior to imputation. After cleaning, imputation, and encoding, the stratified 70:30 split produced 280 training observations and 120 test observations.

The resulting training set was imbalanced (175 CKD vs. 105 non-CKD), which was corrected using SMOTE, as shown in Figure 4.1.

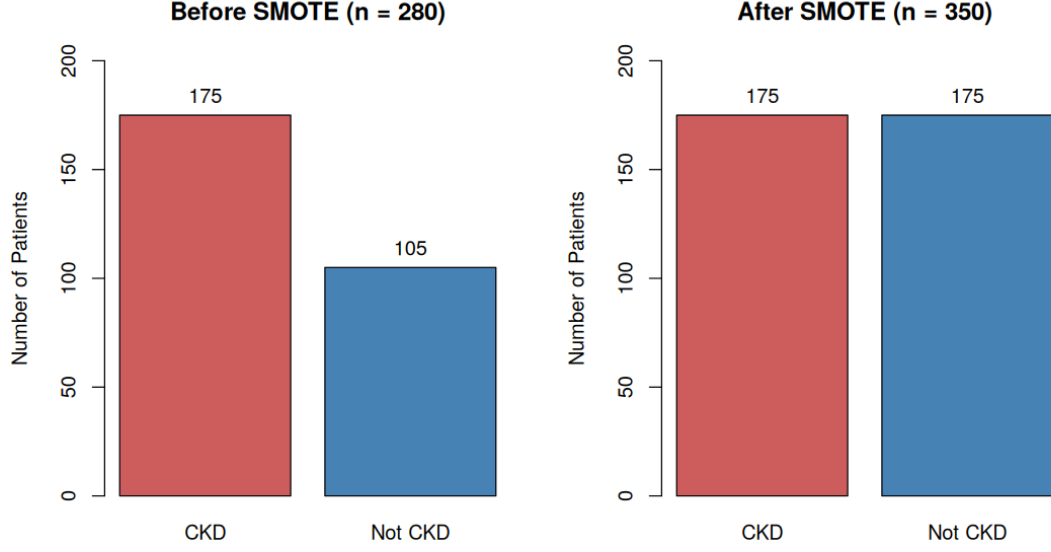


Figure 4.1: Class Distribution Before and After SMOTE (Training Set)

4.3 Classification Results

This section presents the results of the three classification algorithms: RF, LR, and XGBoost, trained on the preprocessed and balanced dataset described in Section 4.2. Each model was evaluated on the same test set using Accuracy, Precision, Recall (Sensitivity), Specificity, F1 Score, and Area Under the Curve (AUC), as defined in Section 3.6.

4.3.1 Random Forest

RF model was trained on the balanced training set described in Section 4.2. The model was built using 70 trees, with 4 variables randomly selected as candidates at each split, corresponding to the floor of the square root of the total number of predictors. The training process required approximately 0.02 seconds, and the resulting model occupied 0.2 MB of memory.

Table 4.1: Out-of-Bag (OOB) Confusion Matrix of RF (Training Set)

	0	1	Class Error
0	98.86%	1.14%	0.011
1	0.57%	99.43%	0.006

Table 4.1 shows the internal Out-of-Bag (OOB) validation results obtained during training. Only 2 negative cases were misclassified as positive, and 1 positive case was misclassified as negative, yielding an OOB error rate of only 0.86% and reflecting strong internal consistency across both classes. This OOB estimate is obtained without a separate validation set, since each tree is evaluated only on the observations left out of its own bootstrap sample.

Table 4.2: Confusion Matrix of RF (Test Set)

	Actual Non-CKD	Actual CKD
Predicted Non-CKD	36.67%	0%
Predicted CKD	0.83%	62.50%

Table 4.2 shows that 44 cases were correctly classified as non-CKD and 75 cases were correctly classified as CKD. Only one non-CKD case was misclassified as CKD, while no CKD cases were misclassified as non-CKD. The closeness between the OOB error rate (0.86%) obtained during training and the error rate observed on the unseen test set (0.83%) indicates that the model generalizes well and shows no evidence of overfitting.

Table 4.3: Performance Metrics of RF

Metric	Value
Accuracy	99.17%
Precision	98.68%
Recall (Sensitivity)	100.00%
Specificity	97.78%
F1 Score	99.33%
Area Under the Curve (AUC)	0.9994

Based on the confusion matrix, RF model achieved excellent predictive performance across all evaluation metrics. The model reached an accuracy of 99.17%,

with a precision of 98.68%, Sensitivity of 100.00%, Specificity of 97.78%, F1 Score of 99.33%, and an AUC of 0.9994.

ROC Curve

Figure 4.2 presents ROC curve of the RF model. The curve rises sharply toward the upper-left corner, indicating excellent discrimination between CKD and non-CKD patients. The model achieved an AUC value of 0.9994, confirming its outstanding predictive performance.

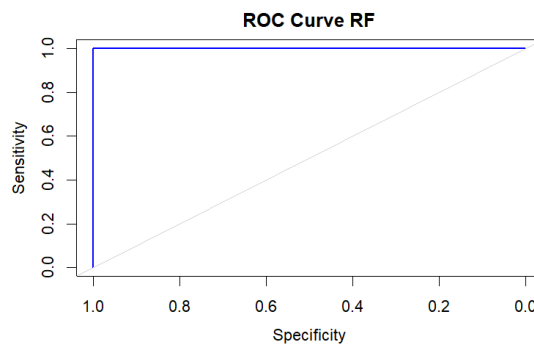


Figure 4.2: ROC Curve of RF

Figure 4.3 presents the variable importance results based on the two criteria defined in Section 3.5.1: mean decrease accuracy and mean decrease Gini. Both criteria consistently rank hemoglobin, serum creatinine, specific gravity, packed cell volume, and red blood cell count among the top contributors to the model, while coronary artery disease, pus cell clumps, and bacteria contributed almost no discriminatory power to either the model's predictive accuracy or its node purity.

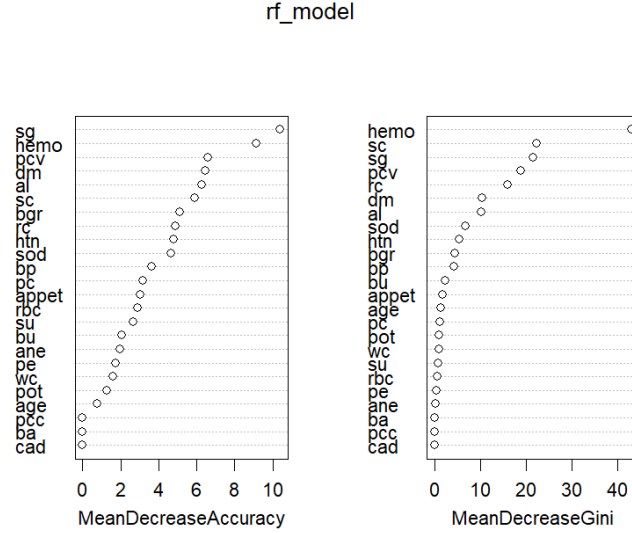


Figure 4.3: Variable Importance of RF (Mean Decrease Accuracy and Mean Decrease Gini)

4.3.2 Logistic Regression

LR model was trained using the balanced training dataset obtained after applying the Synthetic Minority Oversampling Technique (SMOTE). The original dataset was divided into 70% training data (280 observations) and 30% testing data (120 observations). The model was trained using LASSO regularization with the optimal lambda value selected through 10-fold cross-validation. The training process required approximately 0.1112182 seconds, and the memory usage during model construction was 185.0 MB.

Table 4.4: Confusion Matrix of LR (Test Set) — Percentage of Total

	Actual Non-CKD	Actual CKD
Predicted Non-CKD	37.5%	0.83%
Predicted CKD	0%	61.67%

Table ?? shows that 45 cases were correctly classified as non-CKD and 74 cases were correctly classified as CKD. Only one CKD case was incorrectly classified as non-CKD, while no non-CKD cases were misclassified as CKD. These results indicate excellent classification performance on the testing dataset.

Based on the confusion matrix, LR model achieved excellent predictive perfor-

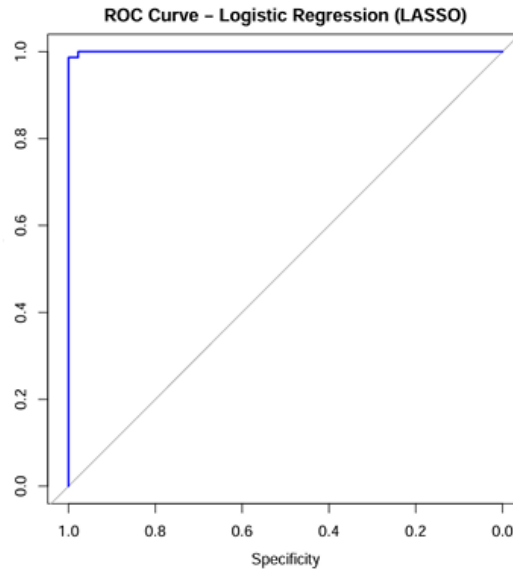
Table 4.5: Performance Metrics of LR

Metric	Value
Accuracy	99.17%
Precision	100.00%
Recall (Sensitivity)	98.67%
Specificity	100.00%
F1 Score	99.33%
Area Under the Curve (AUC)	0.9997

mance across all evaluation metrics. The model reached an accuracy of 99.17%, with a precision of 100.00%, sensitivity of 98.67%, specificity of 100.00%, F1 Score of 99.33%, and an AUC of 0.9997.

ROC Curve

Figure 4.4 Presents ROC curve of LR model. The curve approaches the upper-left corner, indicating excellent discrimination between CKD and non-CKD patients. The model achieved an AUC value of 0.9997, confirming its outstanding predictive performance.

**Figure 4.4:** ROC Curve of LR

The Odds Ratios (ORs) were calculated by exponentiating the estimated regression coefficients. Variables with ORs greater than one increased the likelihood

of CKD, whereas values below one indicated a negative association with the disease. Among the predictors, hypertension, diabetes mellitus, pedal edema, and albumin showed relatively higher Odds Ratios, suggesting stronger associations with CKD. Conversely, variables such as hemoglobin, packed cell volume, and red blood cell count exhibited Odds Ratios below one, reflecting their protective relationship against CKD. The coefficient of (sg) was numerically unstable because of quasi-complete separation; therefore, its Odds Ratio was not interpreted.

Table 4.6: Odds Ratios of LR

Variable	Odds Ratio
dm	26.464
pe	14.152
htn	10.801
al	4.194
rbc	1.592
su	1.167
sc	1.165
bgr	1.018
age	0.999
sod	0.930
pcv	0.757
rc	0.729
hemo	0.385
appet	0.031
sg	0.000

In Table 4.6, dm, pe, and htn were associated with increased odds of CKD, while appet and hemo showed lower odds of CKD. Variables with odds ratios greater than one indicate increased odds, whereas values below one indicate lower odds.

4.3.3 XGBoost

The XGBoost model was trained on the balanced training set, using the configuration specified in Section 3.5.3: a maximum tree depth of 4, a learning rate of 0.1, regularization parameters $\gamma = 1$ and $\lambda = 1$, and 100 boosting rounds with early stopping applied after 10 consecutive rounds without improvement. The

training process required approximately 0.87 seconds, and the resulting model occupied 0.008 MB of memory.

Table 4.7: Confusion Matrix of XGBoost (Test Set)

	Actual Non-CKD	Actual CKD
Predicted Non-CKD	97.78%	0%
Predicted CKD	2.22%	100%

Table 4.7 shows that 44 cases were correctly classified as non-CKD and 75 cases were correctly classified as CKD. Only one non-CKD case was misclassified as CKD, while no CKD cases were misclassified as non-CKD, indicating excellent classification performance on the testing dataset.

Table 4.8: Performance Metrics of XGBoost

Metric	Value
Accuracy	99.17%
Precision	98.68%
Recall (Sensitivity)	100.00%
Specificity	97.78%
F1 Score	99.33%
Area Under the Curve (AUC)	0.9997

Based on the confusion matrix, the XGBoost model achieved excellent predictive performance across all evaluation metrics. The model reached an accuracy of 99.17%, with a precision of 98.68%, sensitivity of 100.00%, specificity of 97.78%, F1 Score of 99.33%, and an Area Under the Curve (AUC) of 0.9997.

ROC Curve

Figure 4.5 presents the Receiver Operating Characteristic (ROC) curve of the XGBoost model. The curve rises sharply toward the upper-left corner, indicating excellent discrimination between CKD and non-CKD patients. The model achieved an AUC value of 0.9997, confirming its outstanding predictive performance.

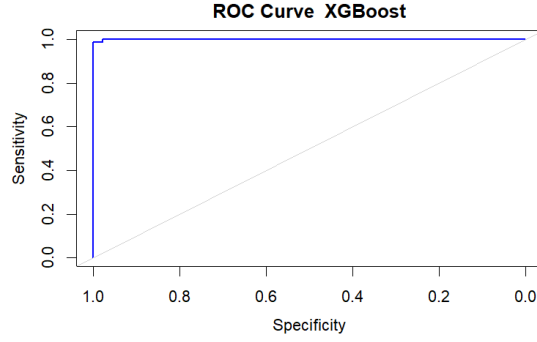


Figure 4.5: ROC Curve of XGBoost

Feature importance was assessed using the Gain criterion, which measures the average improvement in model accuracy contributed by each variable across all splits in which it was used.(Figure 4.7 and Table 4.9)

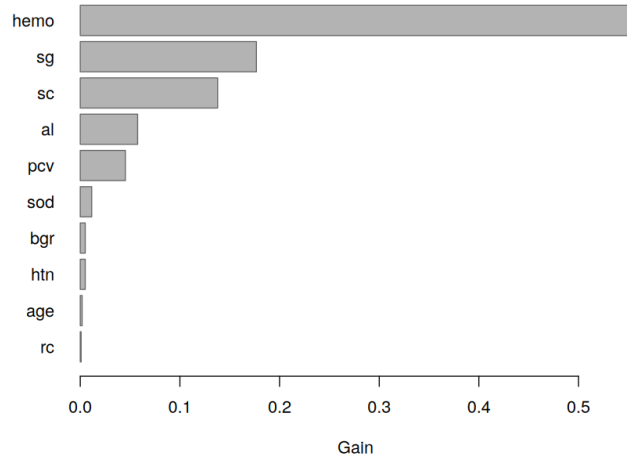


Figure 4.6: Top Features Contributing to the XGBoost Model, Ranked by Gain

Table 4.9: XGBoost Feature Importance (Top 10 by Gain)

Feature	Gain	Cover	Frequency
hemo	0.5580	0.3037	0.2303
sg	0.1768	0.2171	0.2364
sc	0.1379	0.1573	0.1394
al	0.0576	0.1029	0.1394
pcv	0.0453	0.0819	0.0485
sod	0.0115	0.0861	0.0788
htn	0.0050	0.0131	0.0364
bgr	0.0050	0.0204	0.0606
age	0.0019	0.0152	0.0182
rc	0.0009	0.0025	0.0121

In table 4.9 hemoglobin (hemo) was by far the most influential predictor (Gain = 0.558), followed by specific gravity (sg) and serum creatinine (sc). This pattern is consistent with clinical expectations, since anemia, abnormal urine specific gravity, and elevated creatinine are well-established indicators of declining kidney function.

4.3.4 Model Comparison

Table 4.10 summarizes the performance of the three classification algorithms — RF, LR, and XGBoost — on the same test set, using the evaluation metrics defined in Section 3.6.

Table 4.10: Performance Comparison of the Three Classification Models (Test Set)

Model	Accuracy	Precision	Recall	Specificity	F1 Score	AUC
Random Forest	99.17%	98.68%	100.00%	97.78%	99.33%	0.9994
Logistic Regression	99.17%	100.00%	98.67%	100.00%	99.33%	0.9997
XGBoost	99.17%	98.68%	100.00%	97.78%	99.33%	0.9997

All three models achieved an identical Accuracy of 99.17% and an identical F1 Score of 99.33% on the test set, with AUC values above 0.999 in every case. RF and XGBoost produced exactly the same confusion matrix (44 true negatives, 1 false positive, 0 false negatives, 75 true positives), giving them identical Precision, Recall, and Specificity, and differing only marginally in AUC (0.9994 versus 0.9997). LR showed the opposite error pattern: it achieved perfect Precision and Specificity (100.00%), misclassifying no non-CKD patients, but its Recall was slightly lower (98.67%), missing one true CKD case.

This distinction is clinically relevant: the errors made by RF and XGBoost were false positives, which are typically resolved through further testing, whereas the error made by LR was a false negative, which carries a higher risk of delaying diagnosis and treatment.

Regarding variable importance, the three models converge on a similar set of key predictors despite relying on different underlying mechanisms. Hemoglobin,

serum creatinine, specific gravity, and packed cell volume ranked among the top predictors in both the RF (Mean Decrease Gini) and XGBoost (Gain) importance measures, while hemoglobin, packed cell volume, and red blood cell count also emerged as protective factors (Odds Ratios below 1) in the LR model, alongside hypertension, diabetes mellitus, and pedal edema as strong risk-increasing factors (Odds Ratios above 1). This convergence across three methodologically distinct algorithms reinforces the clinical relevance of these biomarkers as reliable indicators of CKD status.

4.4 Clustering

After completing data preprocessing and numerical normalization using Standardization, the K-means algorithm was applied to explore the data's underlying structure without relying on predefined labels.

To determine the optimal number of clusters, the Elbow Method was employed by calculating the WCSS for values of K ranging from 2 to 10.

Table 4.11: WCSS Values for K-Means Clustering Across Different Values of K

K	WCSS
2	3453.827
3	3117.887
4	2825.437
5	2652.371
6	2514.231
7	2413.704
8	2338.092
9	2219.388
10	2122.755

The WCSS curve showed a sharp decline between $K = 2$ and $K = 3$, after which the decrease became gradual, indicating that both values represent statistically viable options.

This behaviour in the WCSS curve indicates that the reduction in WCSS beyond

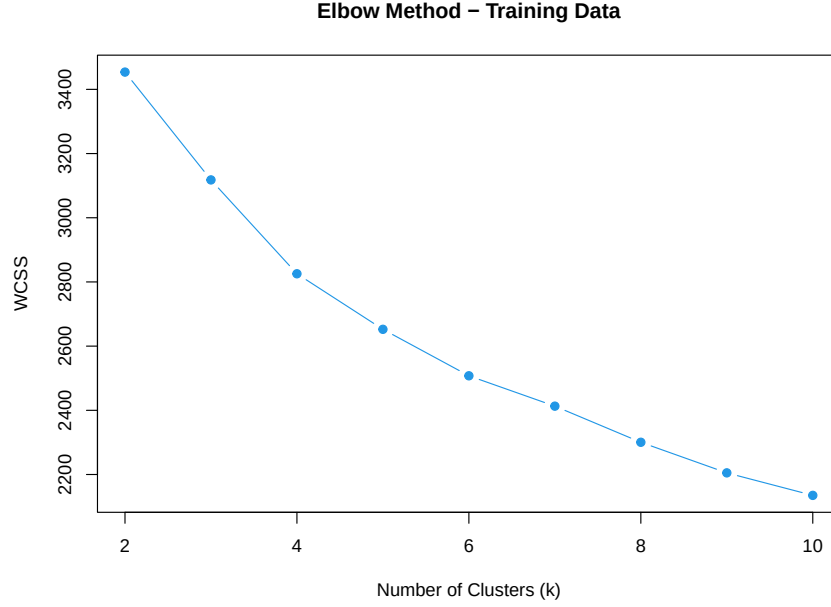


Figure 4.7: Elbow Method for Selecting the Optimal Number of Clusters.

$K = 2$ becomes progressively smaller. Although $K = 3$ shows a further decrease, both values remain statistically defensible choices based on the elbow criterion. To reinforce this decision, the Silhouette Score was calculated for both $K = 2$ and $K = 3$; the value at $K = 2$ was 0.224 and for $K = 3$ it was 0.196. Given that CKD progresses through distinguishable stages of severity rather than a single binary state, the three-cluster solution ($K = 3$) was adopted, as it captures a clinically meaningful stratification into advanced stage, near-normal stage, and Non-CKD groups, while remaining statistically supported by the Elbow Method and an acceptable Silhouette Score.

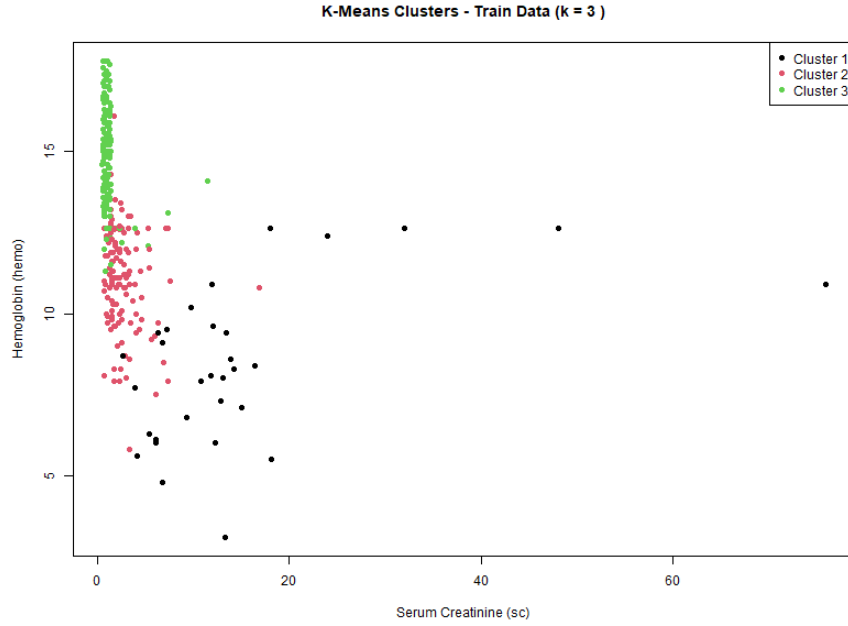


Figure 4.8: K-Means Clusters on the Original Training Data ($K = 3$).

Table 4.12: Cross-Tabulation Between K-Means Clusters and CKD Diagnosis (Original Feature Space)

Cluster	Not CKD (0)	CKD (1)
1	0	31
2	0	145
3	120	24

Clusters 1 and 2 were each composed entirely of CKD patients (100% purity), together capturing 176 of the 200 CKD cases (88%), while Cluster 3 was predominantly composed of non-CKD individuals, containing only a subset of 24 early-stage CKD cases. This indicates that the algorithm was able to isolate the majority of CKD patients into clinically pure subgroups without any prior knowledge of the diagnosis.

When comparing the disease outcomes with the original classification, the results indicate that the model successfully separated a significant proportion of patients without relying on that classification; this suggests that the grouping based on the underlying biological structure effectively identified the relevant biomarkers.

An analysis of the mean biomarker levels for each group was also conducted,

revealing the factors that had the greatest impact on the separation.

Table 4.13: Mean Biomarker Values for Each Cluster

Cluster	sc	hemo	bu	bp
1	14.86	8.37	160.80	81.93
2	2.49	11.30	56.03	80.397
3	1.12	14.82	34.27	71.45

Cluster 1 exhibited the most severe laboratory profile (mean serum creatinine = 14.86, hemoglobin = 8.37), corresponding to an advanced stage subgroup.

Cluster 2 displayed intermediate values (serum creatinine = 2.49, hemoglobin = 11.30), corresponding to a near-normal stage subgroup.

Cluster 3 showed values close to normal ranges (serum creatinine = 1.12, hemoglobin = 14.82), corresponding to a Non-CKD (healthy) subgroup.

This gradient – increasing serum creatinine and blood urea along with decreasing hemoglobin from Cluster 3 to Cluster 2 to Cluster 1 is consistent with the known pathophysiology of chronic kidney disease, in which declining renal function is accompanied by accumulation of metabolic waste and anemia.

4.4.1 What Each Cluster Represents, and the Difference Between Cluster 1 and Cluster 2 (Both CKD)

Although Cluster 1 and Cluster 2 consist solely of patients with chronic kidney disease, they represent two distinct patient populations:

Table 4.14: Clinical Differences Between Cluster 1 (advanced stage) and Cluster 2 (near-normal stage)

	Cluster 1	Cluster 2
<i>n</i>	31	145
sc (Serum creatinine)	14.86	2.49
Hemo (Hemoglobin)	8.37	11.30
bu (Blood urea)	160.80	56.03

Cluster 1 represents a CKD (advanced stage) subgroup, characterized by the highest serum creatinine (SC) and blood urea (BU) values together with the lowest hemoglobin (Hemo) levels, reflecting advanced renal failure.

Cluster 2 represents a CKD (near-normal stage) subgroup, characterized by moderately elevated SC and BU values together with a moderate reduction in Hemo, reflecting chronic kidney disease at a less advanced stage.

Cluster 3 represents the Non-CKD (healthy) subgroup, comprising the majority of non-affected individuals together with 24 cases of early-stage CKD.

Thus,

- Cluster 1 = CKD (advanced stage)
- Cluster 2 = CKD (near-normal stage)
- Cluster 3 = Non-CKD (healthy)

This explains why CKD appeared in two different groups.

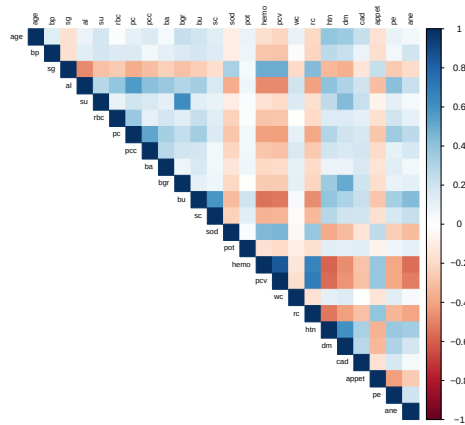


Figure 4.9: Correlation Heatmap of the 14 Continuous Variables

The correlation heatmap shows that several biomarkers move together in predictable patterns. PCA leverages these correlations by combining them into principal components that capture the majority of variance. This transformation reduces noise and emphasizes the underlying CKD severity gradient. It is important to note that the purpose of applying PCA here is not to alter the clustering

outcome or produce a different grouping of patients, but rather to confirm the stability of the three-tier structure identified above while achieving a sharper geometric separation between clusters, as reflected in an improved Silhouette Score.

4.5 Dimensionality Reduction via PCA

Having established a clinically meaningful three-tier severity structure in the original 14-dimensional feature space, PCA was next employed as a complementary analytical step to examine whether this structure remained stable under dimensionality reduction, and whether removing less informative dimensions could sharpen the geometric separation between clusters.

Table 4.15: Variance Explained by Each Principal Component

PC	Standard Deviation	Proportion of Variance	Cumulative Proportion
PC1	2.131	0.324	0.324
PC2	1.289	0.119	0.443
PC3	1.116	0.089	0.532
PC4	0.984	0.069	0.601
PC5	0.959	0.066	0.667
PC6	0.946	0.064	0.731
PC7	0.923	0.061	0.792
PC8	0.813	0.047	0.839
PC9	0.771	0.042	0.881
PC10	0.692	0.034	0.916
PC11	0.625	0.028	0.944
PC12	0.596	0.025	0.969
PC13	0.545	0.021	0.990
PC14	0.372	0.010	1.000

Applying the standard threshold of retaining components that cumulatively explain at least 90% of the total variance, the first ten components (PC1–PC10) were selected, jointly accounting for 91.6% of the total variance in the dataset. This reduced the dimensionality of the clustering input from 14 original variables to 10 uncorrelated components, discarding only the four weakest components (PC11–PC14), which collectively explained a marginal 8.4% of the variance and

were considered to reflect measurement noise rather than clinically meaningful signal.

4.6 K-Means Clustering with Principal Component Analysis (PCA)

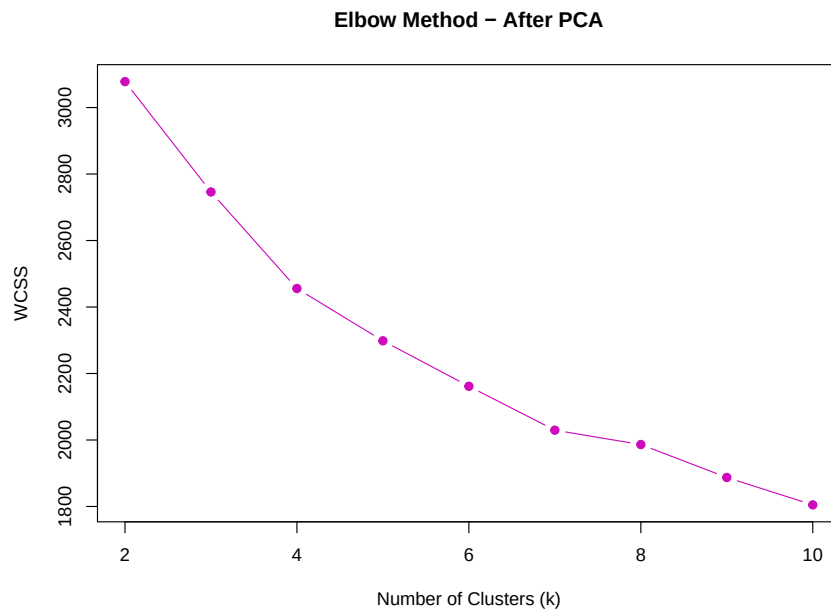


Figure 4.10: Elbow Method After PCA (10 Principal Components)

Applying the Elbow Method to the 10-dimensional PCA-reduced dataset (`x_pca`) produced a WCSS pattern consistent with the original feature space, again showing no single unambiguous elbow point and supporting $k = 3$ as a statistically reasonable choice.

K-Means clustering with $k = 3$ was subsequently fitted to the PCA-reduced data, yielding an average Silhouette Score of 0.216—a modest improvement over the 0.196 obtained on the original 14-variable space.

Cluster-level cohesion varied across the three groups, with silhouette widths of 0.02, 0.39, and 0.08 for Clusters 1, 2, and 3, respectively, indicating that Cluster

2 remained the most internally homogeneous group even after dimensionality reduction.

Table 4.16: Cross-Tabulation Between PCA-Based Clusters and CKD Diagnosis

Cluster	Not CKD (0)	CKD (1)
1	0	31
2	0	144
3	120	25

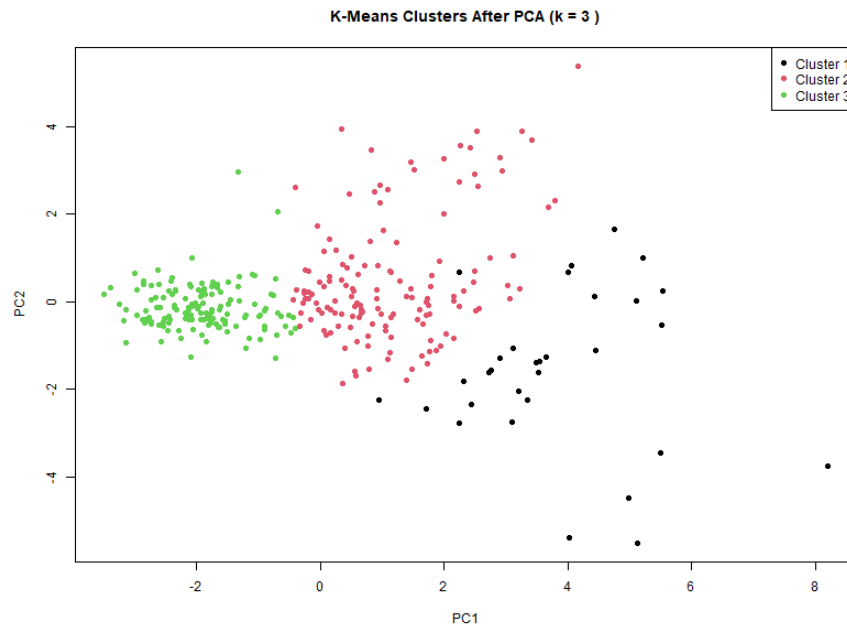


Figure 4.11: K-Means Clusters in PCA Space (PC1 vs PC2)

Without PCA, K-Means ($k = 3$) achieved a Silhouette Score of 0.196, whereas after PCA (10 components) it reached 0.216—a modest but consistent improvement of 0.020. Two related mechanisms explain this:

Noise reduction: The four excluded components (PC11–PC14) collectively explained only 8.4% of the total variance and likely represented random measurement fluctuation rather than meaningful clinical signal. Removing them reduced the “noise” that was previously distorting distance calculations between patients in the original 14-dimensional feature space.

Reduced redundancy: Several original variables were highly correlated (e.g.,

hemoglobin and PCV, $r = 0.88$), meaning they contributed overlapping, partially duplicated information to the distance metric. PCA consolidates this correlated information into fewer, more informative and less redundant axes, which sharpens the geometric boundaries between clusters.

Importantly, this improvement occurred without changing which patients were grouped together (after this) PCA improved the quality of separation, not the underlying grouping itself.

Beyond the modest numerical increase in the Silhouette Score, the practical value of PCA lies in the improved geometric separation between clusters without any meaningful loss in classification accuracy. Comparing the cross-tabulation before PCA (Table 4.12) and after PCA (Table 4.16), the assignment of patients to clusters remained virtually identical, with only a single case reassigned between the original and reduced feature spaces. This indicates that PCA did not alter which patients were grouped together, but rather sharpened how clearly they were separated in space, as reflected in the increased Silhouette Score ($0.196 \rightarrow 0.216$) and in the visibly tighter, more distinct clusters observed in Figure 4.11 compared to the original feature space. In other words, PCA preserved the diagnostic accuracy of the three-tier stratification while enhancing its geometric interpretability.

4.6.1 Comparison Between $k = 2$ and $k = 3$, and Final Decision

Both $K = 2$ and $K = 3$ solutions were evaluated across the original feature space and the PCA-reduced feature space in order to determine the most appropriate number of clusters for this study.

Table 4.17: Comparison of Silhouette Scores for $K = 2$ and $K = 3$ (Original vs PCA Space)

Criterion	$K = 2$	$K = 3$
Silhouette Score (Original Feature Space)	0.224	0.196
Silhouette Score (After PCA)	0.244	0.216

Although the $K = 2$ solution achieved a marginally higher Silhouette Score in both feature spaces (0.224 vs. 0.196 without PCA; 0.244 vs. 0.216 with PCA), the $K = 3$ solution offered substantially higher clinical interpretation, distinguishing patients into high-, moderate-, and low-risk subgroups rather than a single binary distinction.

The three-cluster solution also successfully isolated 176 of the 200 CKD cases (88%) into fully pure clusters, compared to a coarser separation under $K = 2$. Given that the difference in Silhouette Score between $K = 2$ and $K = 3$ was small and that the three-tier structure aligns closely with the known clinical staging of chronic kidney disease, $K = 3$ was adopted as the final clustering solution for this study.

4.6.2 Interpretation of Principal Components (Which Variables Contribute Most to Each PC)

Each principal component represents a new linear direction that integrates highly correlated variables into a single component. To understand the clinical meaning behind each principal component, the loading coefficients (contribution weights) of the original 14 variables were examined for the first three components, which together explained 53.2% of the total variance.

Table 4.18: Principal Component Loadings and Variance Explained (PC1–PC3)

Component	Variance (%)	Cumulative (%)	Dominant Variables (loading)
PC1	32.4	32.4	pcv (0.41), hemo (0.40), rc (0.36), sg (0.31)
PC2	11.9	44.3	su (0.52), bgr (0.48), sc (−0.31), wc (0.30)
PC3	8.9	53.2	age (0.40), pot (0.36), sc (0.33), su (0.31)

Regarding the (+/-) sign, a value (such as `sc` = −0.305 for PC2) does not imply a weaker contribution. Rather, it indicates that this variable moves in the opposite direction to the remaining variables within that principal component. This means that as PC2 increases (indicating higher sugar levels), serum creatinine (SC) tends to decrease, following the mathematical relationship defined by that component.

4.6.3 Spatial Position of Each Cluster in the PCA Plot (PC1 vs. PC2)

Table 4.19: Spatial Position of Clusters in PCA Space (PC1 vs PC2)

Cluster	Position	PC1	Interpretation
Cluster 1 (black) Advanced	Far right	Strong positive	Reflecting the highest biochemical severity (<code>sc</code> = 14.86).
Cluster 2 (pink) Near-normal	Center-left	Near to zero	Mildly elevated, reflecting the near-normal CKD stage (<code>sc</code> = 2.49).
Cluster 3 (green) Non-CKD	Far left	Strong negative	Closest to normal, reflecting the healthy/Non-CKD subgroup (<code>sc</code> = 1.12).

So, in this way, the plot illustrates the separation of groups as a gradient ranging from low to medium to high. The higher the PC1 (on the right), the greater the severity of the disease.

4.6.4 Why the Final Classification Tables (With and Without PCA) Show Nearly Identical Results

The cross-tabulation of clusters against the true diagnosis was nearly identical before PCA (Table 4.12) and after PCA (Table 4.16), differing by only a single reassigned case. This near-identical outcome is explained by the fact that the excluded components (PC11–PC14) accounted for only a very small fraction of the total variance (8.4%), and therefore had negligible influence on which cluster each patient was assigned to.

Rather than being a limitation, this consistency provides evidence for the robustness of the identified cluster structure. It demonstrates that the three-tier risk stratification is not an artifact of a specific feature representation, but instead reflects a genuine and stable pattern in the underlying clinical data. The value of applying PCA was therefore not to alter the grouping outcome, but to confirm its stability while achieving a clearer geometric separation between clusters, as demonstrated by the improved Silhouette Score and the PCA visualization.

Chapter 5

Summary and Future Work

5.1 Introduction

This chapter concludes the study by summarizing the work carried out in the previous chapters, highlighting the most important results, translating these results into practical recommendations, and outlining directions for future research.

5.2 Summary of the Study

This study applied both supervised and unsupervised machine learning techniques to analyze chronic kidney disease (CKD) using a dataset of 400 patients and 24 clinical variables. After extensive preprocessing—including data cleaning, handling missing values, encoding categorical variables, balancing the training set using SMOTE, and standardizing numerical features—the dataset was used to train three supervised classification models: LR, RF, and XGBoost were trained on the balanced training data and evaluated on the untouched test set using Accuracy, Precision, Recall, Specificity, F1-score, and AUC. In parallel, an unsupervised analysis was carried out using K-Means clustering, supported by the Elbow Method, Silhouette Score, and Principal Component Analysis (PCA), to

investigate whether natural severity-based subgroups exist within the CKD population.

Classification performance: All three models demonstrated exceptional and highly comparable performance on the test set, achieving identical Accuracy (99.17%) and F1-scores (99.33%), alongside AUC values exceeding 0.999 across all cases (Table 4.10).

RF and XGBoost yielded identical confusion matrices, each incurring a single false-positive error. Conversely, LR presented the opposite error profile—a single false-negative—thereby failing to identify one true CKD case while maintaining zero false-positive errors.

This distinction is clinically significant: whereas a false positive merely prompts secondary confirmatory testing, a false negative introduces the risk of delayed diagnostic intervention. Consequently, the error distribution of Random Forest and XGBoost is arguably superior for early screening purposes, despite the mathematical parity among the overall metrics.

K-Means clustering ($k = 3$) revealed a clinically meaningful three-tier structure that extends beyond the binary CKD label, categorizing patients into Non-CKD (healthy), CKD (near-normal stage), and CKD (advanced stage) groups. This structural division is well-supported by the elbow method, silhouette analysis, and clear biomarker gradients.

Furthermore, integrating Principal Component Analysis (PCA) to reduce the 14 continuous variables to 10 principal components (explaining 91.6% of the total variance) yielded a modest yet consistent improvement in cluster quality, with the Average Silhouette Score increasing from 0.196 to 0.216, while preserving the underlying patient groupings. This stability confirms that the three-tier severity hierarchy represents a robust, intrinsic feature of the dataset rather than an artifact of the initial feature space, demonstrating that PCA effectively enhanced geometric separation and mitigated redundancy without altering the overarching

clinical insights.

5.3 Conclusions and Recommendations

Hemoglobin, serum creatinine, specific gravity, and packed cell volume should be prioritized in early screening protocols, given their consistent identification as top predictors across all three modeling approaches and their alignment with established clinical understanding of renal function decline.

All three algorithms are viable for CKD screening. Given their near-identical performance, model choice can reasonably be guided by secondary considerations such as interpretability (favoring LR), robustness to nonlinear relationships (favoring RF), or scalability to larger datasets (favoring XGBoost).

A three-tier risk stratification (low, moderate, high) is clinically actionable and could support clinical prioritization decisions, complementing the traditional binary CKD diagnosis by helping clinicians identify patients at high risk of rapid progression.

5.4 Future Work

Building on the findings of this study, several directions for future research are suggested:

1. Expanding feature sets to include genetic, lifestyle, or imaging data for more comprehensive prediction.
2. Investigating temporal models (e.g., survival analysis or time-series approaches) to predict CKD progression rather than static diagnosis.
3. Developing a clinical decision-support tool that incorporates both classification and risk-stratification outputs.

4. External validation using datasets from different hospitals or populations to assess generalizability.

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ملخص البحث

يُعد مرض الكلى المزمن من الأمراض التي قد تتطور دون ظهور أعراض واضحة في مراحلها المبكرة، مما يجعل الكشف المبكر عنه أمرًا مهمًا للحد من حدوث أضرار دائمة في وظائف الكلى. يهدف هذا البحث إلى بناء ومقارنة عدد من نماذج تعلم الآلة لتصنيف المرضى إلى مصابين بمرض الكلى المزمن أو غير مصابين، بالإضافة إلى تحديد أهم العوامل السريرية المرتبطة بالمرض.

استخدمت الدراسة مجموعة بيانات تضم ٤٠٠ مريض و٥٢ متغيرًا سريريًا. وتمت معالجة البيانات من خلال تنظيفها، ومعالجة القيم المفقودة، وترميز المتغيرات الفئوية، ثم تقسيم البيانات إلى مجموعتي تدريب واختبار. كما تم تطبيق تقنية توليد العينات الاصطناعية للفئة الأقل تمثيلًا على بيانات التدريب لمعالجة عدم توازن الفئات.

تم تدريب ومقارنة ثلاثة نماذج للتصنيف، وهي الانحدار اللوجستي، والغابة العشوائية، وخوارزمية إكس جي بوست. وتم تقييم أداء هذه النماذج باستخدام عدة مقاييس، من أهمها الدقة، والاستدعاء، ودرجة، والمساحة تحت منحنى خاصية تشغيل المستقبل. كما تم استخدام خوارزمية المتوسطات وتحليل المكونات الرئيسية للكشف عن مجموعات كامنة بين المرضى تتجاوز التصنيف الثنائي للمرض.

أظهرت النتائج أداءً متقاربًا وممتازًا للنماذج الثلاثة، حيث حقق كل منها دقة بلغت ٩٩,١٧ في المئة مع قيمة للمساحة تحت المنحنى تجاوزت ٩٩,٠. كما أظهرت النتائج أن الهيموغلوبين، وحجم الخلايا المرصوفة، والكثافة النوعية، وكرياتينين المصل، والألبومين كانت من أهم المتغيرات المرتبطة بالمرض. كذلك أسفر تحليل المتوسطات عن تحديد ثلاث مجموعات للمرضى تمثل مستويات خطورة مرتفعة ومتوسطة ومنخفضة، ودعم تحليل المكونات الرئيسية استقرار هذا التقسيم.



التنبؤ بمرض الكلى المزمن وتصنيفه باستخدام تقنيات تعلم الآلة

إعداد:

إسراء بادويل ٢٣٠٧١٢٦

بلقيس العلق ٢٣١١٨١٢

نوره النفيعي ٢١١٣٣٨٩

وسن السميري ٢٢٠٩٤٠٤

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